

An Empirical Overview of Stress Detection Based on Photoplethysmography Signal Using Various Deep Learning Techniques

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Abstract—The rapid speed of modern life and the shift towards cognitive work have contributed to a rising prevalence of mental stress across all professions. Many sedentary occupations, particularly in fields like Information Technology (IT), causes prolonged periods of intense mental effort that can significantly elevate stress levels. Extended exposure to such mental stress is linked to an increased risk of serious conditions, such as cardiovascular diseases and mental health disorders. Consequently, there is a critical need for non-intrusive tools capable of detecting mental stress. Early detection is essential for preventing stress-related health concerns and continuous observation of stress levels. Numerous methods are currently available for stress detection, but they face limitations like inaccuracies due to individual differences and environmental factors as well as difficulties with real-time monitoring. In this discussion, several techniques are explored that offer better insights with stress detection, providing valuable perspectives for future researchers. This survey discusses various deep learning techniques, highlighting their advantages and limitations. It also suggests potential improvements to tackle these limitations in the future. Initially, this survey categorizes Deep Learning techniques into five techniques: Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), Transfer Learning (TL), Attention-Based Mechanisms, and Transformer-based approaches are discussed in depth. Subsequently, it identifies research gaps in traditional Deep Learning approaches to guide future innovative model development. The study reviews 25 research papers focusing on various methods for stress detection using DL. Finally, this survey evaluates performance metrics, datasets, techniques used for research, and model achievements. CNNs emerge as the predominant approach, and Python is frequently used for stress detection in DL. Additionally, accuracy is highlighted as a key analytical metric.

Keywords: *Convolutional Neural Network (CNN), Long-Short Term Memory (LSTM), Photoplethysmogram (PPG), Transfer Learning (TL), Deep Learning (DL).*

I. INTRODUCTION

Photo plethysmography (PPG) is a non-invasive, cost-effective method employed to measure blood volume and heart rate in patients. This technique involves a semiconductor device that illuminates skin tissue and captures variations in light intensity and volume [1]. The PPG signal is produced by directing a 660-1100 nm wavelength Light-Emitting Diode (LED) through the subject's index

finger, while a photodiode on the opposite side detects light absorption during each cardiac cycle [2]. The resulting PPG signal contains valuable information, including latent psycho physiological data, and captures the arterial pulse wave created by the heart's contractions and reflections. [3]. Stress is a physical and mental reaction that individuals experience when confronted with challenging or unfamiliar situations [4]. Stress can be detected through various methods, including physiological monitoring, behavioral analysis, self-reported surveys, Deep Learning (DL), and Machine Learning (ML). Among these, deep learning is particularly effective in identifying stress.

Recently, the Deep learning (DL) is gaining vast attention for its capability to understand intricate patterns and representations using enormous datasets. DL models, such as CNNs, LSTM, Graph Neural Networks (GNNs), Generative Adversarial Networks (GANs), transfer learning, attention-based mechanisms, and Transformer-based models, are capable of extracting valuable features and achieving precise predictions [5]. Convolutional neural networks (CNNs) are predominantly utilized in computer vision applications and bio-signal classification tasks. For instance, they are employed in tasks like Electromyogram (EMG) signal classification for gesture recognition and Electroencephalogram (EEG) pattern identification for assistive machine control [6]. Long Short-Term Memory (LSTM) networks are characterized by memory blocks within their recurrent connections, enabling them to retain information over extended periods. LSTMs have demonstrated effectiveness in processing medical signals like Electrocardiogram (ECG), underscoring the potential of this network architecture in healthcare applications [7]. DL models excel in automatically extracting intricate features from PPG data, enabling accurate classification of stress levels [8]. Preprocessing [9], feature extraction [10], feature selection [11], and stress detection [12] [13] are steps used in the stress detection methods.

The modules employed for detection of stress via PPG in DL is catalogued into CNN, LSTM, TL, Attention based mechanisms and Transformer based models. This survey explores a range of DL techniques, detailing their specific methods used, pros and cons and provides an in-depth

analysis of how these techniques perform in various applications and identifies key areas where they reveal gaps. Additionally, the survey details potential strategies and improvements to address these limitations, aiming to enhance the effectiveness and applicability of deep learning in future research and practice. This review analyzes the tools used, databases utilized, classification of approaches, and metrics applied.

II. LITERATURE REVIEW

This segment elaborates a survey of various approaches for stress detection using PPG within DL. Fig. 1 represents different types of models involved in detection of stress using PPG signals in DL.

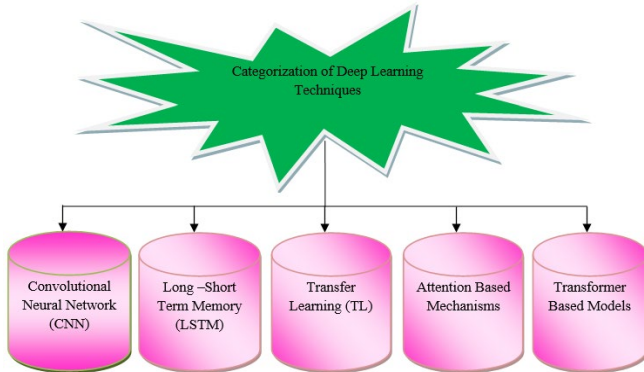


Fig. 1 Categorization based on different approaches

A. CNN based approach

A powerful deep learning model named the Convolutional Neural Network (CNN) intended primarily for computer vision tasks like image classification. By processing input images, CNNs provide outputs that indicate the class or the possibility of an image belonging to a particular category. CNNs mitigate issues, such as overfitting and effectively address limitations inherent in those methods. Due to their superior performance, CNNs are widely applied across various domains to tackle complex challenges such as image classification and object detection.

De Pedro-Carracedo, J., *et al.* [8] developed a binary classification model for analyzing the stress level that uses the convolutional neural network (CNN) and works well of dynamic nature of PPG signals. The PPG signal's robust diffusive dynamics against both internal and external artifacts allow for precise identification of intense stress episodes with minimal signal sampling, achieving high accuracy and sensitivity. However, despite using a single PPG signal, it lacked a comprehensive understanding of the physiological mechanisms associated with stress, which limited effective and systematic optimization of the CNN model through hyperparameter adjustments.

Barki, H. and Chung, W.Y., [14] devised an ear mounted biosensor based on 2D-CNN that uses PPG signals for stress detection. Initially, the PPG signal was processed and transformed into scalograms using Continuous Wavelet Transform (CWT), followed by the use of a CNN classifier to distinguish between stressed and non-stressed states. Though this method achieved a higher accuracy, a constraint

of this method was the comparatively limited sampling size that potentially limits the applicability to a wider population.

Chakraborty, S., *et al.* [15] proposed a Multichannel Convolutional Neural Network design for identifying states of mind using bio signals from wearable devices. Recognizing that a single bio signal is insufficient for this task, they utilized multiple signals along with the Adam Optimizer for improved optimization. While this approach reduced computational costs, it fell short in accurately identifying the state of mind with various bio signals.

Elzeiny, S. and Qaraqe, M., [16] devised a stress level identification system using the CNN-based method that is efficient in analyzing the spatial and frequency domain representations from the PPG signal. This model achieved a high accuracy but the limitation lies in that the models developed here focus on classifying individuals into only two categories as stress or non-stress, based on instant recognition of stress using physiological data. Predicting stress over time was not addressed within the scope of this research.

Fontes, L., *et al.* [12] introduced a novel deep learning technique based on remote photoplethysmography (rPPG), incorporating 1D Convolutional Neural Network (1D-CNN), LSTM, and Gated Recurrent Units (GRU) models. This model significantly boosts both accuracy and efficiency in detecting stress using the UBFC-Phys dataset and maintained tremendous computational performance. However, it ignored the probable impact of applicant ethnicity on accuracy, highlighting the significance to address diversity within the dataset and its implication for extended application of the stress detection framework.

Elzeiny, S. and Qaraqe, M., [13] also proposed a wearable model for stressor identification using CNN, categorizing images of extracted Interbeat Intervals (IBI) and Blood Volume Pulse (BVP) from PPG sensors into stress states (stressed or non-stressed) and stressor types (physical, cognitive, social) in both spatial and frequency domains. This method outperformed existing techniques but was limited by the volume of data collected, which affected overall performance.

Rescio, G., *et al.* [17] developed three neural networks—1D-CNN, LSTM, and GRU—for effective mental stress detection. This model ensures reliability even during sensor malfunctions and provides versatile and effective monitoring under various scenarios. However, it did not explore additional classification techniques, particularly neglecting unsupervised methods.

Onim, M.S.H., *et al.* [18] presented a Random Forest and a Fully Connected Network with CNN model for stress level identification that validates the integration of digital stress biomarkers and provides deeper insights into stress responses without relying on self-reported labels. The limitation of this model was that it only had a narrow focus on classification techniques and inadequate exploration of unsupervised learning methods, alongside challenges with contextual variability and data quality.

Eom, S., *et al.* [19] introduced SIM-CNN, a self-supervised learning method for stress recognition using multimodal biosignals. This model performed unlabeled data efficiently, enhancing model robustness and generalization across diverse datasets and environments without the need for extensive labeled data. However, the complexity and computational resources required for training multimodal models, which limits its scalability in large-scale deployment.

B. LSTM based approach

LSTM networks are specialized RNNs that have the ability of learning and retaining long-term dependency in sequential data, make them well-suited for tasks such as speech recognition and analysis of time series data. LSTMs are utilized in stress detection by analyzing physiological signals over time, effectively capturing subtle variations in data patterns that correlate with stress levels, making them suitable for real-time monitoring and analysis.

Zhu, H., *et al.* [3] developed an attentive deep recurrent architecture called Arousal-valence Networks (ArvaNets) that relies on graph convolutional networks and recurrent neural networks. Also, a LSTM is considered with sequential factor to determine the mental states of the individual. Although this method was effective in determining stress, it failed due to the unstability factor in data driven tasks.

Ziaratnia, S., *et al.* [7] combined two deep learning approaches—Compact Convolutional Transformer (CCT) and Long Short-Term Memory (LSTM)—into a CCT-LSTM pipeline. A CCT model was used to extract features from each modality, followed by an LSTM block for temporal pattern recognition. However, this method lacked an in-depth analysis of hyperparameters, such as window size and moving interval, during the data preprocessing stage, impacting system efficiency.

Rastgoo, M.N., [20] created a CNN and LSTM approach with vehicle and contextual data to jointly learn correlated representations, effectively combining multiple sensory inputs to improve real-time driver stress level classification accuracy and robustness. Anyhow, the potential complexity and computational demands of integrating multiple types of sensory data in real-time applications limit the feasibility of deploying the system in resource-constrained environments or systems.

Masuda, N *et al.* [21] introduced a Multi-Input CNN-LSTM classification model combining CNN and LSTM to accurately classify the stress levels. This method enhanced accuracy through combined EEG and physiological signal analysis, and robustness in capturing both temporal dynamics and spatial features, optimizing performance across diverse fear-inducing scenarios. However, classification of stress levels has potential challenges, such as the complexity of model training and interpretation, which demands substantial computational resources and specialized expertise for effective implementation and optimization.

C. TL based approach

Transfer learning leverages pre-trained models to enhance performance on new but related tasks, reducing training time and data requirements. It helps to improve model generalization and is especially useful when dealing with limited data. It is widely applied in image recognition, natural language processing, and medical diagnostics.

Li, F., *et al.* [22] proposed a novel method using pulse rate variability and elastic-net regression for predicting mental stress through deep learning techniques. Their results indicated high prediction accuracy and stability with field datasets, showcasing the model's strong ecological adaptability and robustness. However, it did not assess chronic stress, which is essential for accurately representing an individual's stress levels.

Radhika, K. *et al.* [23] introduced a transfer learning system for subject-independent stress detection using physiological signals, enhancing generalization across subjects, reducing data collection efforts, and improving model robustness to inter-individual variability. However, it had limitations like potential biases from pre-training on specific datasets, difficulty in transferring knowledge across diverse populations due to physiological variability, and challenges in ensuring privacy and ethical handling of sensitive physiological data.

D. Attention based Mechanisms

Attention-based mechanisms focus on relevant parts of the input, dynamically weighting information to improve performance and context understanding. They enhance model interpretability, manage long sequences effectively, and increase computational efficiency.

Lee, H., Chang, J., *et al.* [24] developed an attention based deep Neural Network Mechanism by integrating raw signals and human-engineered features and achieved better performance, boosts accuracy, yielding state-of-the-art results across different modalities in stress classification. Anyhow, it did not succeed in integrating a broader array of bio signals, including ECG, and utilizing larger, more diverse datasets to enhance the models' ability to achieve higher accuracy.

Coca, M.D.P., *et al.* [11] developed an attention-based system for stress identification, focusing on morphological parameters of PPG signals, which effectively detected two states: rest and attention tasks. This method overlooked potential complementary insights from integrating both photoplethysmography waveform analysis and heart rate variability, which could provide a more comprehensive understanding of low stress states during attention tests but the method was incapable to deliver better insights due to its small sample size and possible inaccuracies in PPG and heart rate variability measurements.

El-Hajj, C. *et al.* [25] created attention-based recurrent neural networks for blood pressure estimation using PPG sensors. This model enhanced accuracy in BP estimation by focusing on informative PPG signal segments, and reduced cost and complexity with single-sensor implementation,

facilitating convenient and real-time monitoring in diverse healthcare settings. But it faced limitations, such as potential challenges in robustness to diverse physiological conditions and variability in PPG signals across individuals, requiring validation across broader populations and conditions to ensure reliable BP estimation.

E. Transformer based Models

Transformer-based models excel at capturing long-range dependencies and contextual relationships in data, enabling efficient parallel processing that accelerates training and inference. Their scalability allows them to handle large datasets and complex tasks effectively. Additionally, they are versatile, achieving state-of-the-art performance across various domains like text, image, and speech processing.

Xu, M., *et al.* [26] designed a transformer-based architecture tailored to pulsative signal structures, featuring a Bottleneck Dilated Convolutional (BDC) transformer to leverage the quasiperiodicity inherent in the data. However, its limitation lies in the ECG signal interference in fMRI environments and the corrupted data of seismocardiograms with movement disturbances.

Kuzmanov, I., *et al.* [27] developed various transformer models specifically for processing biological signals, focusing on transformers that analyze regularly sampled biological recordings from the human body. These models faced challenges including data inefficiency due to limited data availability, interpretability issues in complex biological contexts, and adaptation constraints to capture specific domain characteristics effectively.

Barki, H., *et al.* [28] presented an innovative in-ear PPG device for diagnosing mental stress. However, its limitations include reliance on a narrow dataset, focusing solely on in-ear plethysmography which may miss other relevant signals, and potential challenges with scalability and interpretability of Vision Transformer models.

F. Others

Other methods such as Artificial Intelligence and Machine Learning are used in identifying stress in health care systems, but Deep Learning is particularly notable for its enhanced effectiveness in stress detection. Some of the techniques are covered below.

Paul, A., *et al.* [29] presented a unique approach based on multimodal characterization using PPG signal to identify the stress levels. This technique used the data from the DEAP data set for identification. This model effectively detect stress by integrating multimodal data with PPG signals, improving accuracy and personalization for healthcare applications. However, it failed to address the limitations associated with integrating multiple modalities for mental stress detection using PPG signals in personal healthcare applications, potentially limiting its practical applicability or generalizability.

Salankar, N., *et al.* [10] introduced an effective method for detecting stress markers in different brain lobes by

processing multimodal physiological signals, utilizing Variational Mode Decomposition (VMD) for data preprocessing and decomposing signals into oscillatory mode functions. However, this method failed to classify other neurological disorders, such as epilepsy, depression and Alzheimer's disease.

Albertetti, F., *et al.* [4] introduced a wrist-device sensor to capture physiological signals indicative of stress. However, its limitation is that it must tackle the challenges related to power consumption, connectivity, and user acceptance.

Mitro, N., *et al.* [30] developed an AI-enabled smart wristband for monitoring passengers' physiological states and stress during critical situations, such as large passenger ship evacuations. This approach effectively tracks vital signs and stress levels, providing timely health management insights. However, it faces limitations concerning data privacy and security due to the sensitive nature of health information, as well as potential inaccuracies in stress assessments from physiological signals, necessitating validation and calibration for reliable application in diverse contexts. Table I shows the methods, advantages, and disadvantages of each method.

III. ASSESSMENT AND DISCUSSION

This section examines various studies on stress detection using PPG signals in DL. This survey categorizes modules, explores publication years, and discusses metrics used for estimation.

A. Evaluation with approaches

This section discusses various deep learning approaches for the detection of stress using PPG signals, as outlined in Fig. 2. CNN techniques were employed in 36% of the research papers, while 16% utilized LSTM models. Additionally, Transfer Learning was utilized in 8% of the models and Attention-based mechanisms were used in 12%. Furthermore, Transformer-based models were employed in 12% of the papers, with the remaining 16% based on other DL techniques. In this, CNNs are widely used in DL as it has the capability of extracting hierarchical feature automatically them effective for image and spatial data analysis.

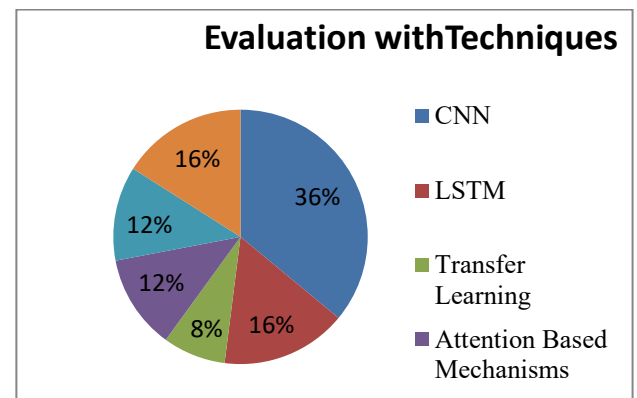


Fig. 2 Evaluation based on techniques for Stress detection in DL

B. Estimation on basis of toolset

This section examines the tools used in previous approaches for stress detection in DL, which is illustrated in Fig. 3. The evaluated tools include PYTHON, Kubios, MATLAB, and SCANerTM. PYTHON emerged as the predominant simulation tool used in the majority of research papers for stress detection in DL due to its extensive libraries and frameworks, such as TensorFlow, Keras, and PyTorch, which simplify the development and implementation of the models also, the readability, flexibility, and strong community support make it an ideal language for rapid experimentation and research.

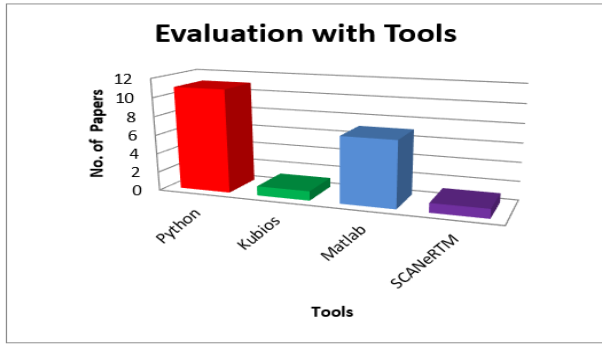


Fig. 3 Evaluation with tools for Stress detection in DL

C. Assessment regarding dataset

This section explores the datasets employed in prior approaches for stress detection using deep learning, as illustrated in Fig. 4. The datasets encompass a variety of datasets including Real-time dataset, WESAD, UBFC – Phys, DEAP, EEGMAT, DESY, CLAS, Nature Scientific dataset, MIMIC II dataset, Laboratorial dataset, and PPG-Dalia dataset. Among these studies, WESAD is the most widely used dataset. Its diverse and annotated data make it valuable for training and evaluating models on stress and affect detection tasks.

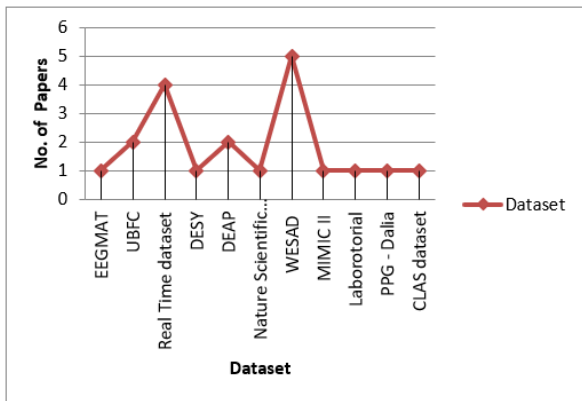


Fig. 4 Evaluation with datasets for Stress detection in DL

D. Assessment on basis of evaluation metrics

This section reviews analytic measures used for stress detection in deep learning. Table I indicates that 22 papers employed accuracy as an evaluation metric, 9 used F1-Score, 4 utilized Precision, 4 applied Recall, 1 adopted AUC, and 1 used Sensitivity as systematic analytical measures.

TABLE I. ASSESSMENT BASED ON EVALUATION METRICS

Evaluation Metrics	No. of Papers
Accuracy	[8] [14] [16] [12] [3] [13] [29] [22] [17] [24] [11] [26] [10] [18] [27] [7] [28] [23] [19] [21] [25] [30]
F1 – Score	[8] [14] [17] [27] [7] [28] [20] [19] [21] [
Precision	[14] [15] [17] [23]
Recall	[14] [15] [17] [23]
AUC	[19]
Sensitivity	[8]

E. Assessment regarding Accuracy

Table II shows the analysis based on accuracy. This ranges from 71% to 75%, 76% to 80%, 86% to 90%, 91% to 95%, and 96% to 100%. Paper [3] achieved the lowest accuracy in the range of 71% to 75%, while research papers [10] and [4] recorded the highest accuracy within the range of 96% to 100%.

TABLE II. ASSESSMENT BASED ON ACCURACY

Accuracy range %	Number of papers
70 - 75 %	[3]
76 – 80 %	[23] [19]
86 – 90 %	[22] [11]
91 – 95 %	[14] [12] [17] [26] [28] [20] [30]
96 – 100 %	[8] [16] [13] [29] [24] [10] [18] [4] [7] [21] [25]

IV. CONCLUSION

This survey outlines the typical framework of stress detection techniques in using PPG signals in DL. It categorizes the scheduling approaches into five main methods: CNN, LSTM, Transfer Learning, Attention-based Mechanisms, and Transformer-based Models, providing detailed discussions on each. It identifies research gaps in modules related to stress detection using PPG signals in DL, highlighting shortcomings overlooked by previous methodologies to guide future researchers in developing innovative models. This study encompasses analysis from 25 research articles that explore various methodologies for stress detection. Among this, CNN is the mostly preferred for stress detection using PPG signals due to their robustness in extracting temporal features and handling noise, enabling efficient and accurate analysis of physiological data. PYTHON tool is used in most of the papers due to its versatility, ease of prototyping by facilitating rapid development and implementation of DL models on diverse datasets including PPG signals. WESAD data set is widely used due to its comprehensive collection of physiological signals, diverse stress-inducing activities, and standardized protocols. Although, CNNs are highly effective in detecting stress future research should focus on optimizing their efficiency and scalability across various applications by integrating them with multimodal data sources advancing interpretability and ethical considerations in AI systems. This analysis will guide researchers to intensify efforts in detecting

stress using DL, with the goal of lowering energy consumption and costs.

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